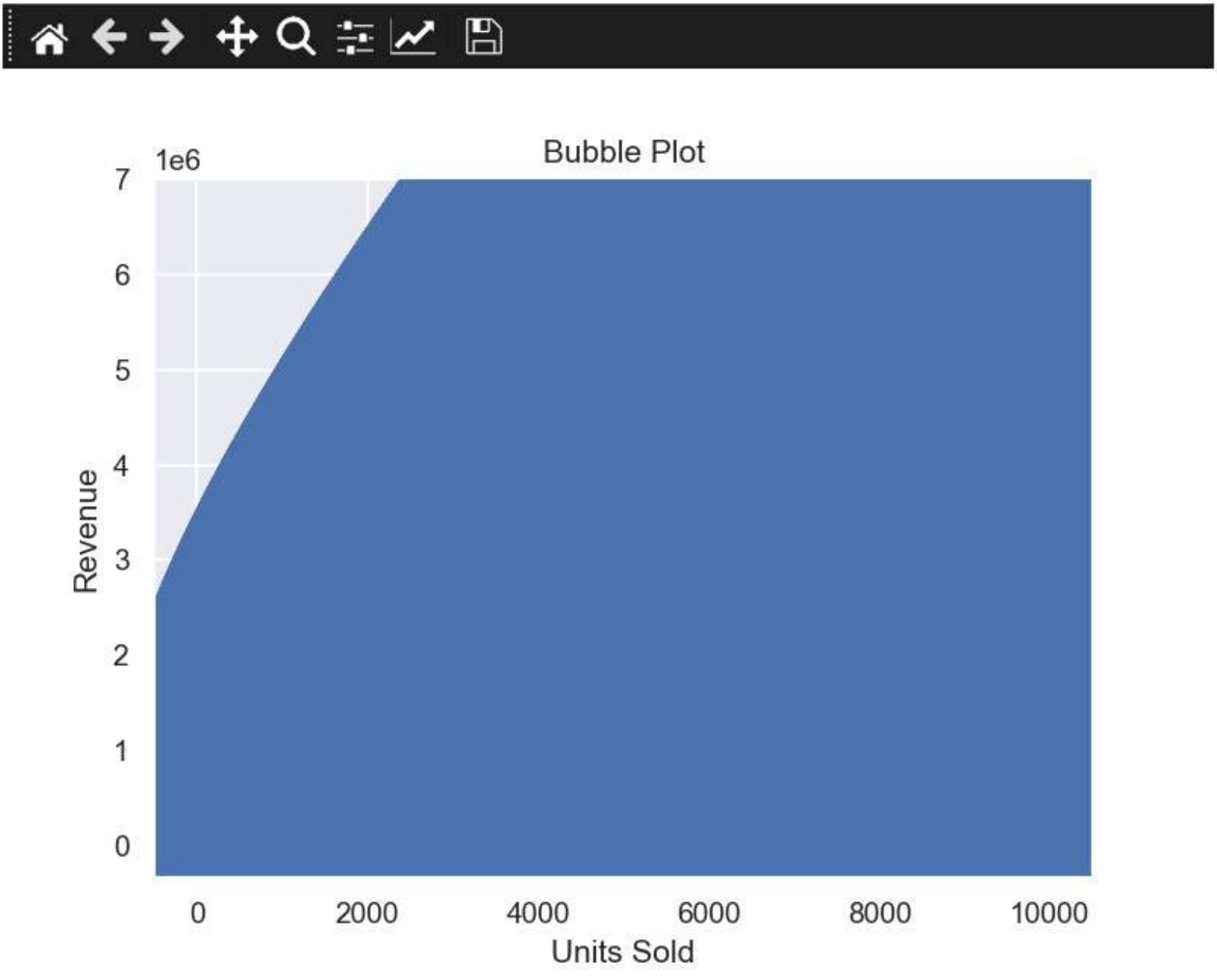
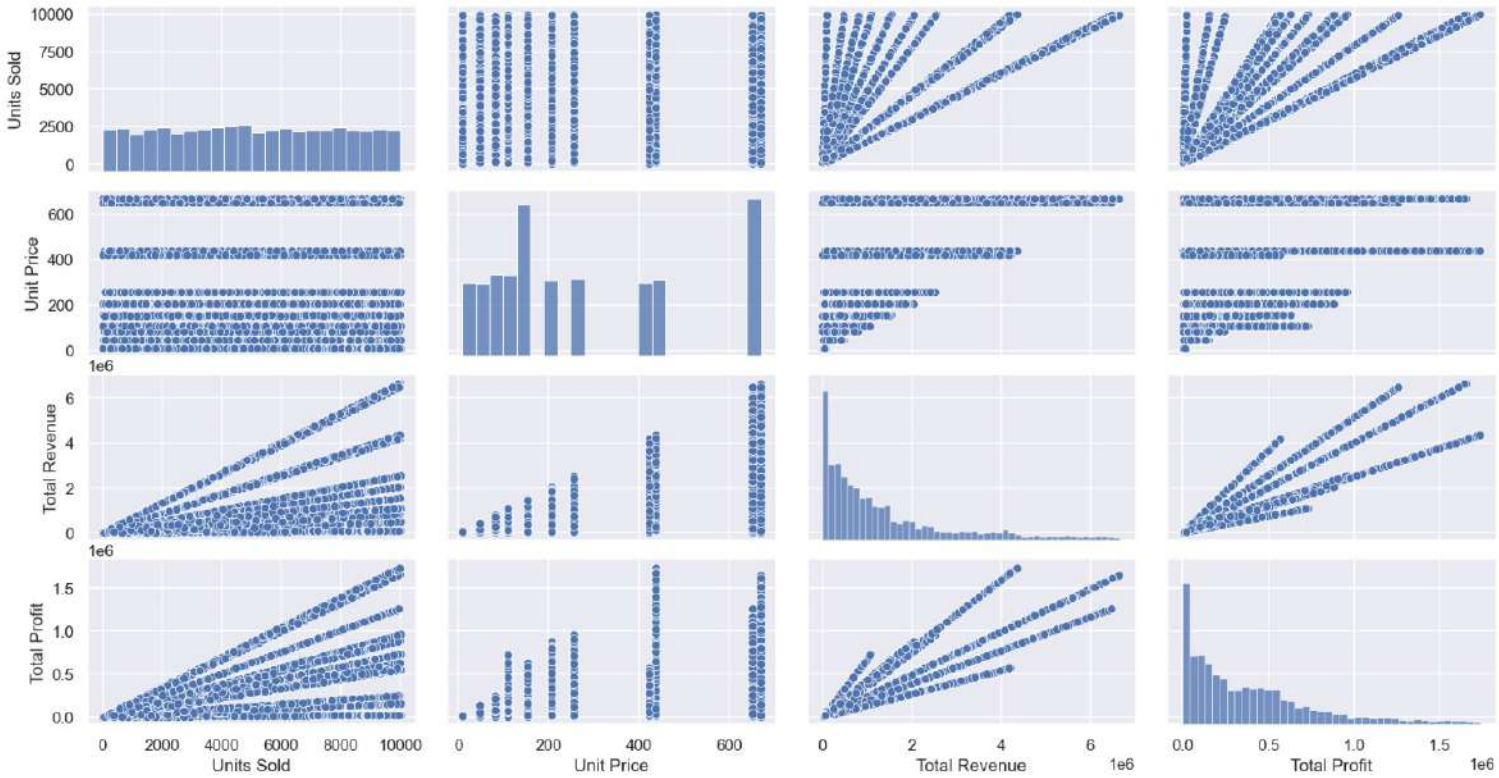
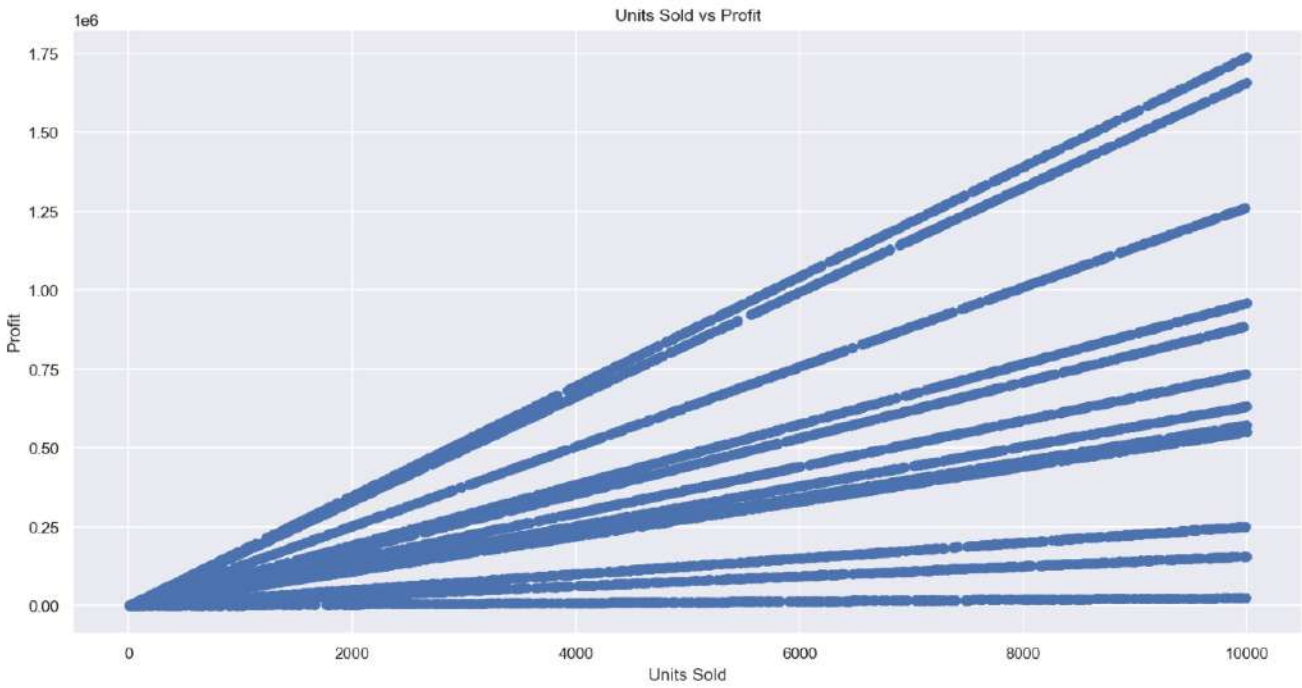
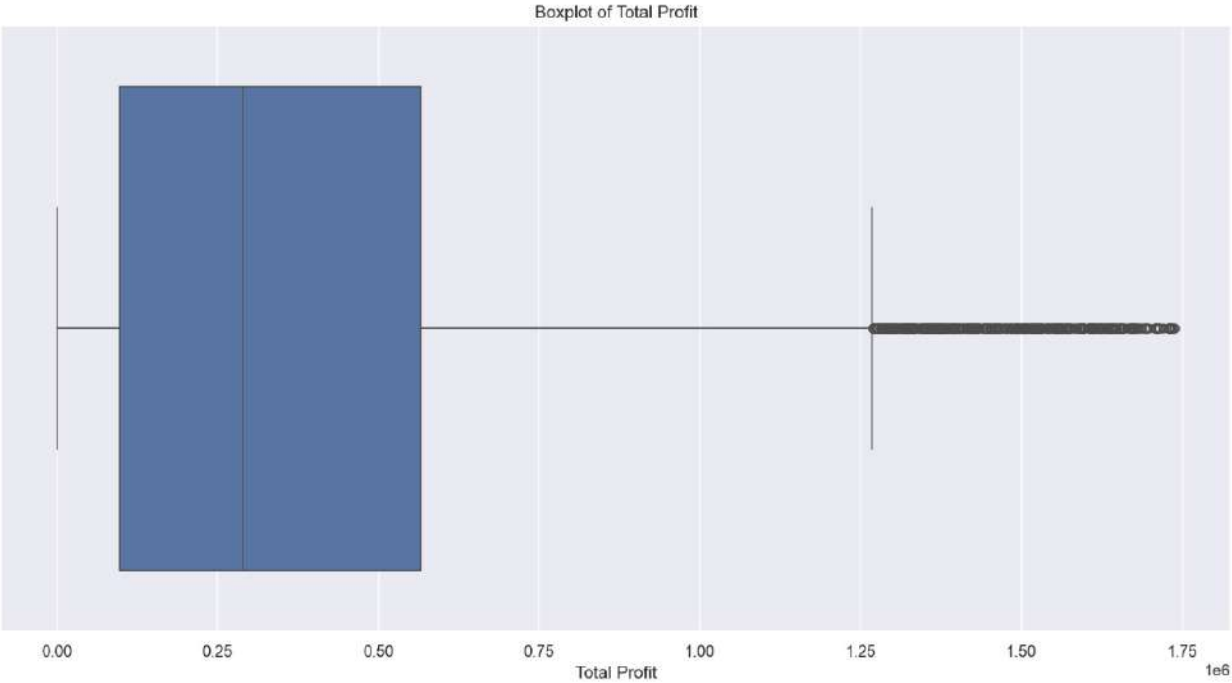
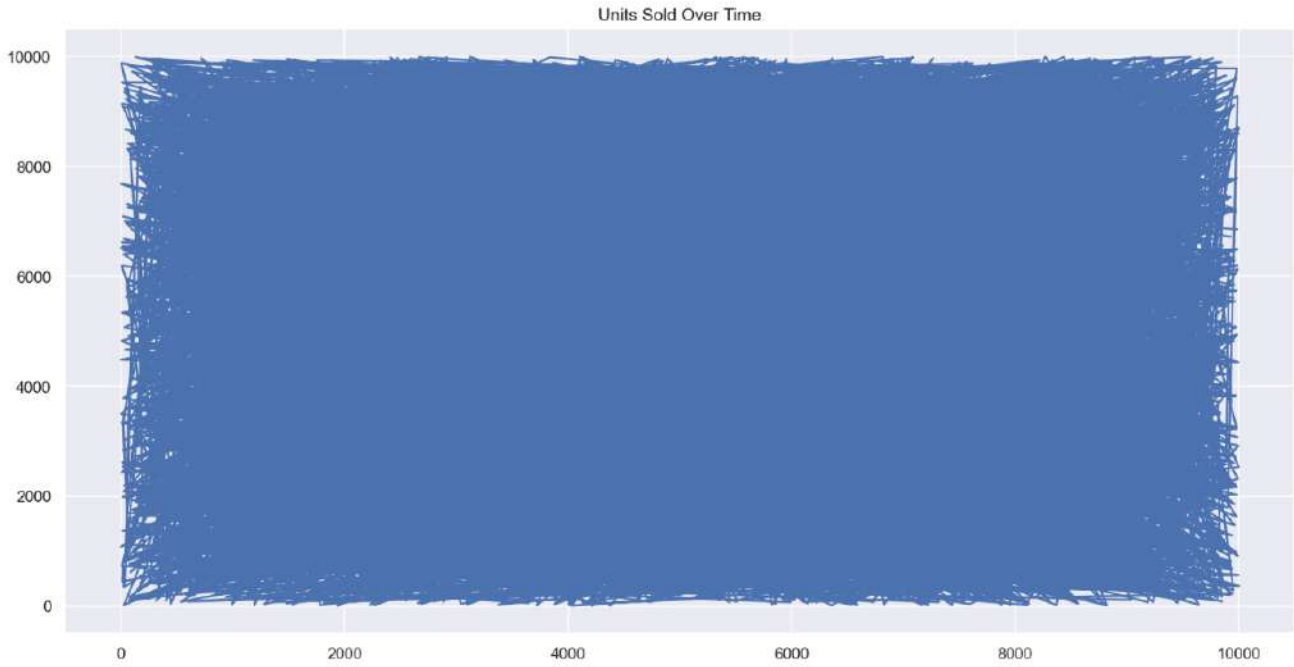
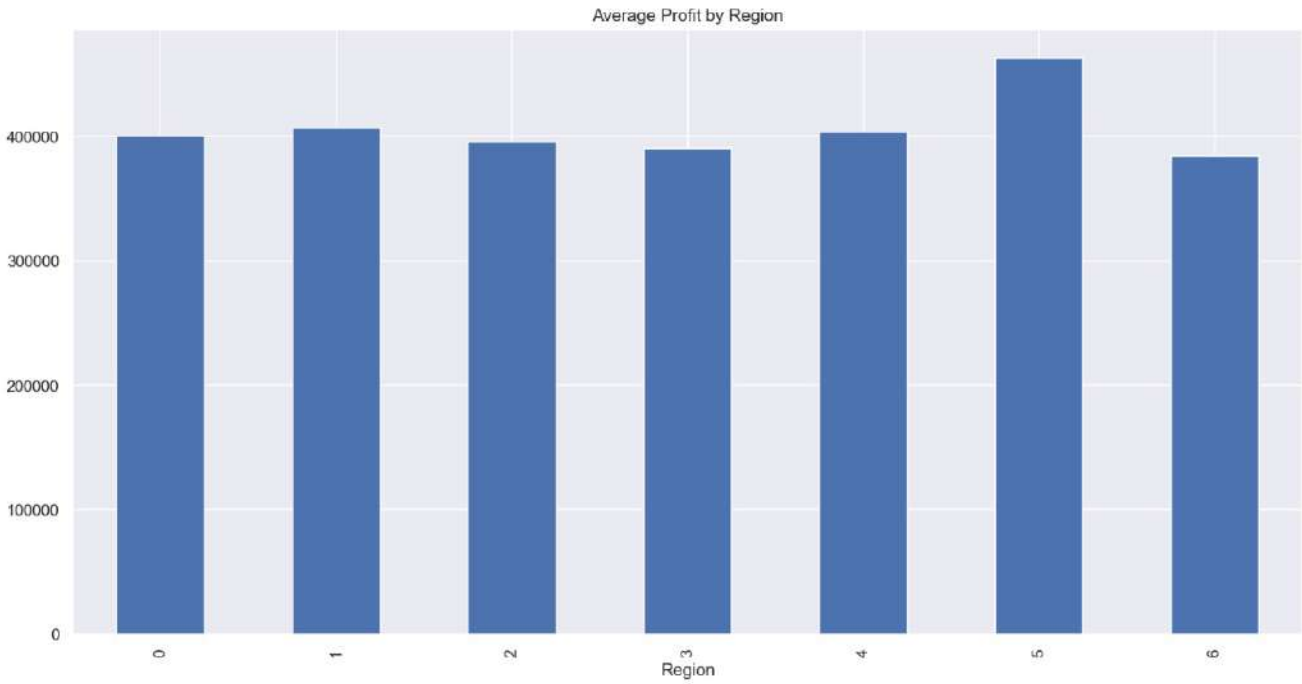

Bubble Plot
1e6
7
6
5
4
3
2
1
0
Revenue
0
2000
4000
6000
8000
10000
Units Sold

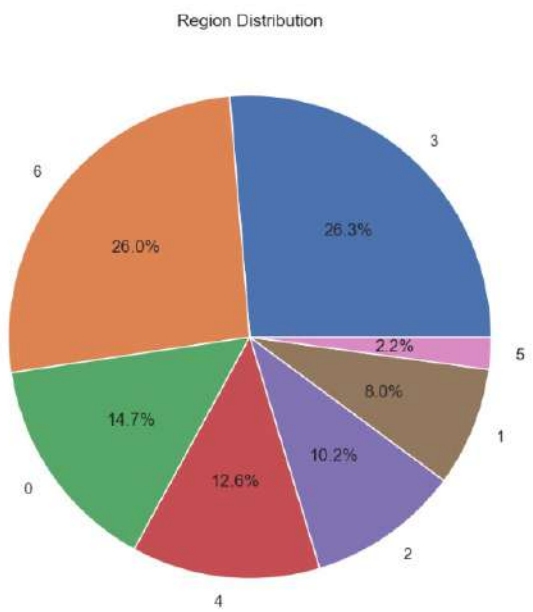

Units Sold
Unit Price
Total Revenue
Total Profit
1e6
1e6
1e6

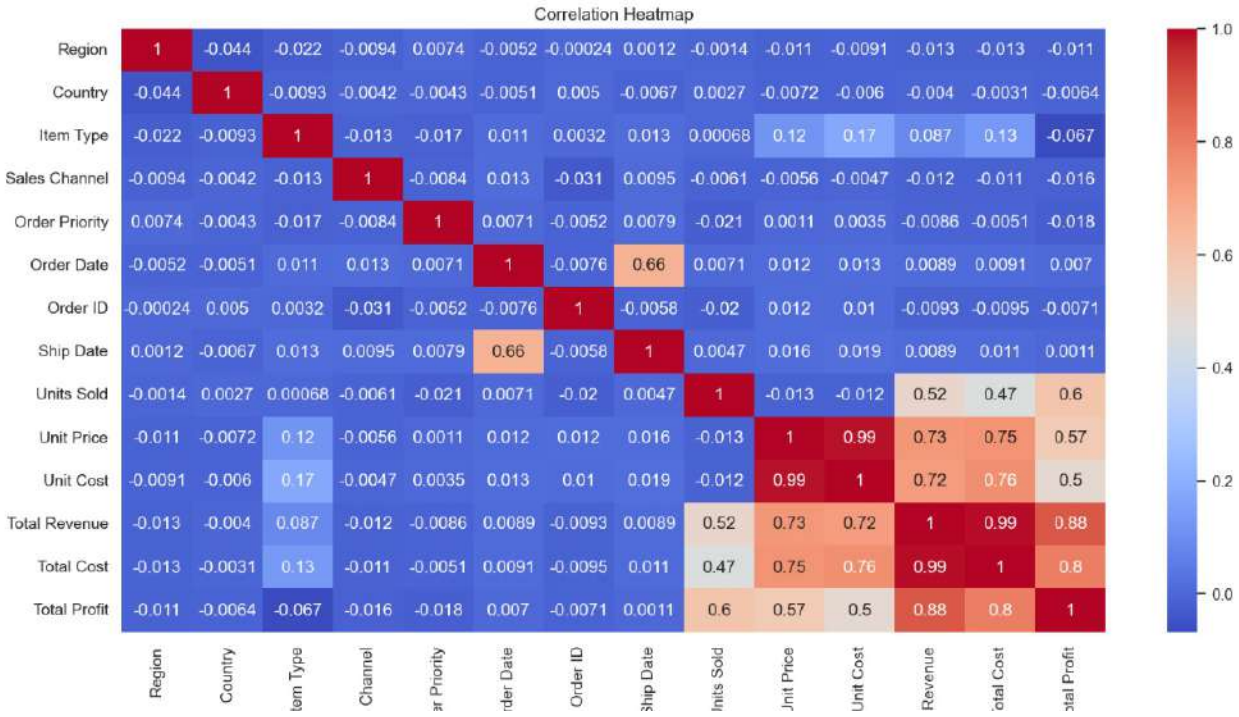

Units Sold vs Profit
1e6
1.75
1.50
1.25
1.00
0.75
0.50
0.25
0.00
Profit
0
2000
4000
6000
8000
10000
Units Sold

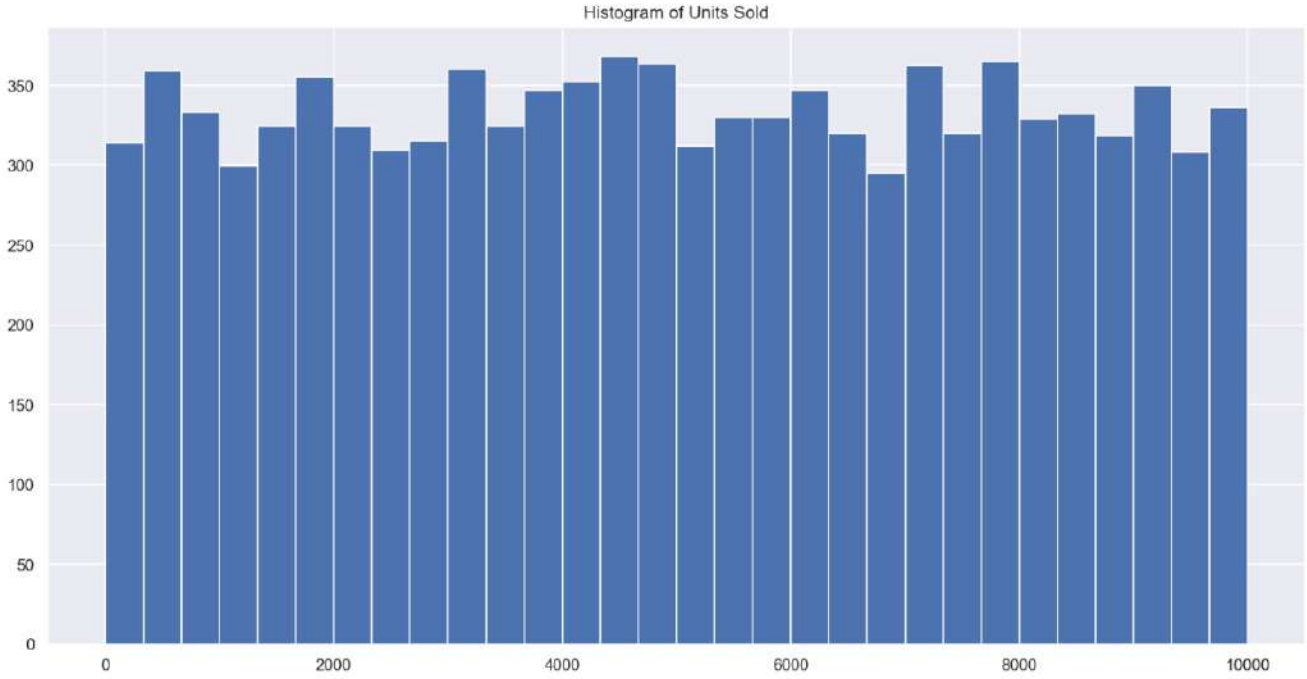

Boxplot of Total Profit
0.00
0.25
0.50
0.75
1.00
1.25
1.50
1.75
1e6
Total Profit


Units Sold Over Time
10000
8000
6000
4000
2000
0
0
2000
4000
6000
8000
10000


Average Profit by Region
400000
300000
200000
100000
0
0
1
2
3
4
5
6
Region


Region Distribution
3
26.3%
6
26.0%
5
2.2%
1
8.0%
2
10.2%
4
12.6%
0
14.7%


Correlation Heatmap
Region 1 -0.044 -0.022 -0.0094 0.0074 -0.0052 -0.00024 0.0012 -0.0014 -0.011 -0.0091 -0.013 -0.013 -0.011
Country -0.044 1 -0.0093 -0.0042 -0.0043 -0.0051 0.005 -0.0067 0.0027 -0.0072 -0.006 -0.004 -0.0031 -0.0064
Item Type -0.022 -0.0093 1 -0.013 -0.017 0.011 0.0032 0.013 0.00068 0.12 0.17 0.087 0.13 -0.067
Sales Channel -0.0094 -0.0042 -0.013 1 -0.0084 0.013 -0.031 0.0095 -0.0061 -0.0056 -0.0047 -0.012 -0.011 -0.016
Order Priority 0.0074 -0.0043 -0.017 -0.0084 1 0.0071 -0.0052 0.0079 -0.021 0.0011 0.0035 -0.0086 -0.0051 -0.018
Order Date -0.0052 -0.0051 0.011 0.013 0.0071 1 -0.0076 0.66 0.0071 0.012 0.013 0.0089 0.0091 0.007
Order ID -0.00024 0.005 0.0032 -0.031 -0.0052 -0.0076 1 -0.0058 -0.02 0.012 0.01 -0.0093 -0.0095 -0.0071
Ship Date 0.0012 -0.0067 0.013 0.0095 0.0079 0.66 -0.0058 1 0.0047 0.016 0.019 0.0089 0.011 0.0011
Units Sold -0.0014 0.0027 0.00068 -0.0061 -0.021 0.0071 -0.02 0.0047 1 -0.013 -0.012 0.52 0.47 0.6
Unit Price -0.011 -0.0072 0.12 -0.0056 0.0011 0.012 0.012 0.016 -0.013 1 0.99 0.73 0.75 0.57
Unit Cost -0.0091 -0.006 0.17 -0.0047 0.0035 0.013 0.01 0.019 -0.012 0.99 1 0.72 0.76 0.5
Total Revenue -0.013 -0.004 0.087 -0.012 -0.0086 0.0089 -0.0093 0.0089 0.52 0.73 0.72 1 0.99 0.88
Total Cost -0.013 -0.0031 0.13 -0.011 -0.0051 0.0091 -0.0095 0.011 0.47 0.75 0.76 0.99 1 0.8
Total Profit -0.011 -0.0064 -0.067 -0.016 -0.018 0.007 -0.0071 0.0011 0.6 0.57 0.5 0.88 0.8 1
Region
Country
tem Type
Channel
r Priority
rder Date
Order ID
hip Date
nits Sold
nit Price
nit Cost
Revenue
otal Cost
otal Profit
1.0
0.8
0.6
0.4
0.2
0.0


Histogram of Units Sold
350
300
250
200
150
100
50
0
0
2000
4000
6000
8000
10000

```
T-Statistic: -0.5820574236608662
P-Value: 0.5605530716555313

Statistical Summary:
             Region       Country  ...    Total Cost  Total Profit
count  10000.000000  10000.000000  ...  1.000000e+04  1.000000e+04
mean       3.248300     92.051600  ...  9.382658e+05  3.950893e+05
std        2.058317     53.384089  ...  1.145914e+06  3.775550e+05
min        0.000000      0.000000  ...  1.245600e+02  4.338000e+01
25%        2.000000     46.000000  ...  1.647855e+05  9.832914e+04
50%        3.000000     91.000000  ...  4.816058e+05  2.890990e+05
75%        6.000000    138.000000  ...  1.183822e+06  5.664227e+05
max        6.000000    184.000000  ...  5.241726e+06  1.738178e+06

[8 rows x 14 columns]

Linear Regression R2 Score: 0.688913624413595
MAE: 164095.58351028548

ML Model R2 Score: 1.0
>>>
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